

A composite transformation is described as a rotation of 60° clockwise with respect to the point $(30, -90)$, followed by a reflection about the line $9x - 9y + 423 = 0$, followed by another differential scaling by a factor of $(2, 0.5)$ with respect to the point $(423, 0)$ and finally followed by a translation of $(0, 423)$. Find the composite transformation matrix for the above transformation. [You do not need to converge the composition into one matrix]

Solution:

- i) Rotation $\rightarrow 60^\circ \rightarrow$ Clockwise w.r.t point $(30, -90)$
- ii) Reflection $\rightarrow$ line $9x - 9y + 423 = 0$
- iii) Scaling $\rightarrow (2, 0.5)$ w.r.t point $(423, 0)$
- iv) Translation $\rightarrow (0, 423)$

- i) Rotation: steps: i) Translate to origin, $T(-30, 90)$
- ii) Rotation, $R(-60)$
- iii) Translate back $T(30, -90)$

$$M_R = T(30, -90) R(-60) T(-30, 90)$$

$$= \begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & -90 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-60^\circ) & -\sin(-60^\circ) & 0 \\ \sin(-60^\circ) & \cos(-60^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -30 \\ 0 & 1 & 90 \\ 0 & 0 & 1 \end{bmatrix}$$

- ii) Reflection: steps: i) Translate to Origin, $T(0, -b)$
- ii) Rotation, $R(-\theta)$
- iii) Reflection about x-axis, R_{fx}
- iv) Rotation, $R(\theta)$
- v) Translate back, $T(0, b)$

$$\begin{aligned}
 9x - 9y + 423 &= 0 & \therefore \tan \theta &= m \\
 \Rightarrow x - y + 47 &= 0 & \Rightarrow \theta &= \tan^{-1}(m) \\
 & & &= \tan^{-1}(1) \\
 & & &= 45^\circ
 \end{aligned}$$

$$\Rightarrow y = \underset{\substack{\uparrow \\ m=1}}{x} + \underset{\substack{\uparrow \\ b=47}}{47}$$

$$M_{Ref} = T(0, b) \times R(\theta) \times R_{fx} \times R(-\theta) \times T(0, -b)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 47 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -47 \\ 0 & 0 & 1 \end{bmatrix}$$

(iii) Scaling: Steps: i) Translate to Origin, $T(-423, 0)$
 ii) Scaling, $S(2, 0.5)$
 iii) Translate back, $T(423, 0)$

$$M_{SC} = T(423, 0) \times S(2, 0.5) \times T(-423, 0)$$

$$= \begin{bmatrix} 1 & 0 & 423 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -423 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(iv) Translation:

$$M_T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 423 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore M = M_T \times M_{SC} \times M_{Ref} \times M_R$$